# NANDINI MEHTA

**SOFTWARE ENGINEERING PORTFOLIO | 2026**

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## Software systems built for reliability, traceability and measurable outcomes.

This portfolio highlights selected work across backend services, full stack applications, data engineering, applied AI, automated testing, CI/CD, monitoring, high volume data workflows and production focused software development.

**01 BACKEND AND APIS**

### Chargeholidays Hotel Booking Platform

**Technologies:** Go · REST APIs · PostgreSQL · Firebase · Elasticsearch · Kibana · Docker · Kubernetes · Google Cloud

**Impact:** Professional backend work at ChargeHorizons on systems processing more than 50,000 events per day.

- Developed Go microservices and REST APIs backed by PostgreSQL and Firebase Firestore.
- Integrated Elasticsearch for hotel search, filtering and availability workflows.
- Built Kibana monitoring for logs, service behaviour and operational visibility.
- Worked with Docker, Kubernetes and CI/CD for automated testing and deployment.

**02 COMMERCE**

### StoreFront E Commerce Platform

**Technologies:** React.js · Node.js · PHP · MySQL · JavaScript · MVC

**Impact:** Built a complete commerce application spanning user flows, backend logic, database operations and testing.

- Implemented a full stack e commerce application with React and Node.js plus a PHP backend.
- Connected product and user workflows to a MySQL data layer.
- Used MVC structure, prototyping and testing cycles to keep application logic maintainable.
- Worked across frontend, backend and database concerns instead of treating them as isolated layers.

**03 TESTING AND QUALITY**

### Python Web Applications and Automated Validation

**Technologies:** Python · Django · Flask · PostgreSQL · Pytest · Docker

**Impact:** Maintained 5 plus applications serving around 2,000 daily users and reduced the QA cycle by 25 percent.

- Built and maintained Python web applications and REST APIs with Django, Flask and PostgreSQL.
- Automated validation using Pytest and repeatable test workflows.
- Improved maintainability through reusable backend components and database integration.
- Used Docker and Git based development practices to support consistent engineering workflows.

## RELIABILITY AND SCALE

04 MONITORING

### Real Time Vehicle Fault and Emission Anomaly Detection Platform

**Technologies:** Python · FastAPI · PostgreSQL · React · WebSocket · Monitoring

**Impact:** Built an end to end telemetry system with structured alerts, operational KPIs and root cause visibility.

- Streamed and validated vehicle telemetry through backend services and PostgreSQL.
- Built live dashboards with React and WebSocket updates.
- Designed alerts, trends and operational indicators for abnormal system behaviour.
- Structured validation and monitoring so failures could be detected and investigated consistently.

05 LARGE DATA

### Euro 7 Automotive Data and ML Pipeline

**Technologies:** Python · Pandas · NumPy · scikit learn · Autoencoders · PCA · UMAP

**Impact:** Processed more than 25 GB and approximately 90 million OBD and CAN time series records.

- Built modular processing steps for cleaning, transformation, feature preparation and validation.
- Developed anomaly detection and clustering workflows for noisy real world engineering data.
- Created repeatable model evaluation and result validation procedures.
- Produced traceable technical findings and test specifications for engineering stakeholders.

06 API AND DATA INTEGRITY

### ClaimVision AI Multimodal Insurance Claim Analyzer

**Technologies:** Python · FastAPI · PostgreSQL · OCR · Structured Extraction · Validation

**Impact:** Designed a modular processing system where multiple input types produce consistent structured outputs.

- Combined claim documents, repair invoices and vehicle damage images in one analysis workflow.
- Added extracted values, consistency checks and risk indicators with validation across inputs.
- Separated backend processing components so they could be tested, debugged and evolved independently.
- Focused on data contracts, traceability and reliable API ready outputs.

07 DATA PROCESSING

### Document Classification and Extraction

**Technologies:** Python · FastAPI · Tesseract · Hugging Face · NLP · Structured Outputs

**Impact:** Built a reusable document ingestion and extraction workflow with validation at the API boundary.

- Processed PDF and image documents through OCR, classification and field extraction steps.
- Added FastAPI interfaces with input validation and structured response formats.
- Designed reusable processing components for invoices, receipts and statements.
- Used repeatable test inputs to debug extraction quality and data consistency.

## CORE ENGINEERING CAPABILITIES

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|---|---|
| **Software engineering** | Go, Python, Java and PHP experience across APIs, microservices, web backends and full stack applications. |
| **Data and databases** | Professional PostgreSQL experience plus MySQL, Oracle, Firebase, Elasticsearch and large scale data processing. |
| **Automated testing** | Unit and integration testing plus Pytest automation that reduced a QA cycle by 25 percent. |
| **Scalable systems** | Services processing more than 50,000 events per day plus a data pipeline over approximately 90 million records. |
| **Messaging and events** | Worked with Google Cloud Pub/Sub, MQTT and event driven application components. |
| **Reliability and observability** | Elasticsearch, Kibana, Grafana, monitoring, debugging, alerts and service health investigation. |
| **Cloud delivery** | Docker, Kubernetes, Google Cloud, Git based workflows and automated CI/CD delivery. |
| **Product development** | Built ecommerce, booking, document intelligence, AI and real time monitoring applications across the stack. |
| **AI and agentic systems** | LLM APIs, RAG, LangGraph, Hugging Face, multimodal AI, OCR, NLP, semantic search, vector databases and model evaluation. |

## ADDITIONAL ENGINEERING PROJECT

### Wildlife Protecting System | Bachelor Thesis

Built an embedded railway safety system using Java, Arduino and PIR sensors with MQTT and HTTP based alerting. The project strengthened my foundations in event driven communication, real time monitoring and system reliability.

## PROFESSIONAL OUTCOMES

**50,000+**

events per day supported by backend and cloud services

**90 million**

automotive records processed in a large data pipeline

**2,000**

daily users across Python web applications

**25 percent**

reduction in QA cycle through Pytest automation

## ENGINEERING STRENGTHS

- Design APIs and software components around clear data contracts and traceable behaviour.
- Treat automated testing, monitoring and CI/CD as part of feature delivery from the beginning.
- Use databases, validation and observability to protect data consistency and diagnose failures quickly.
- Combine backend, full stack, data, AI and agentic systems experience with event driven cloud development.

## CONTACT

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